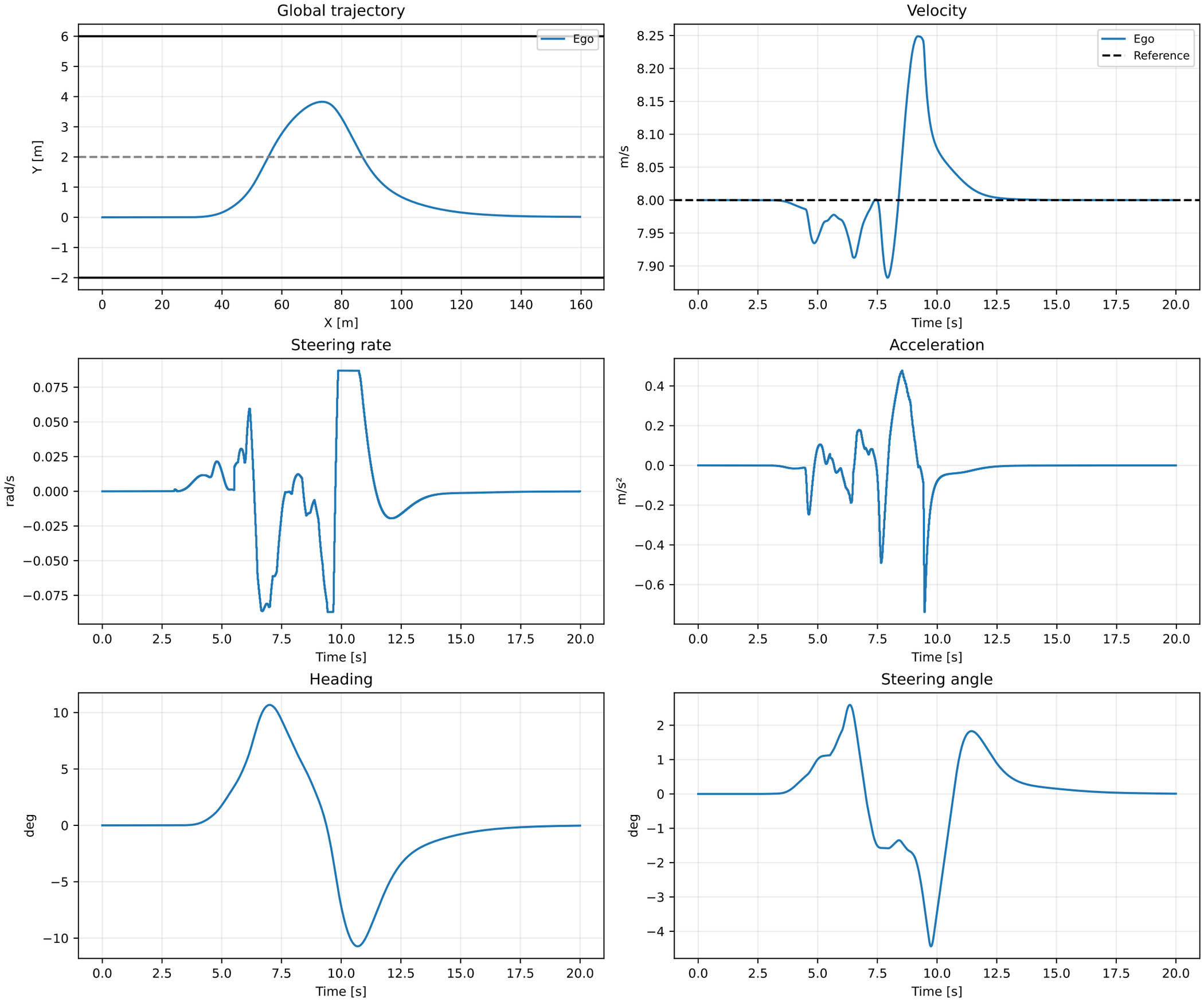
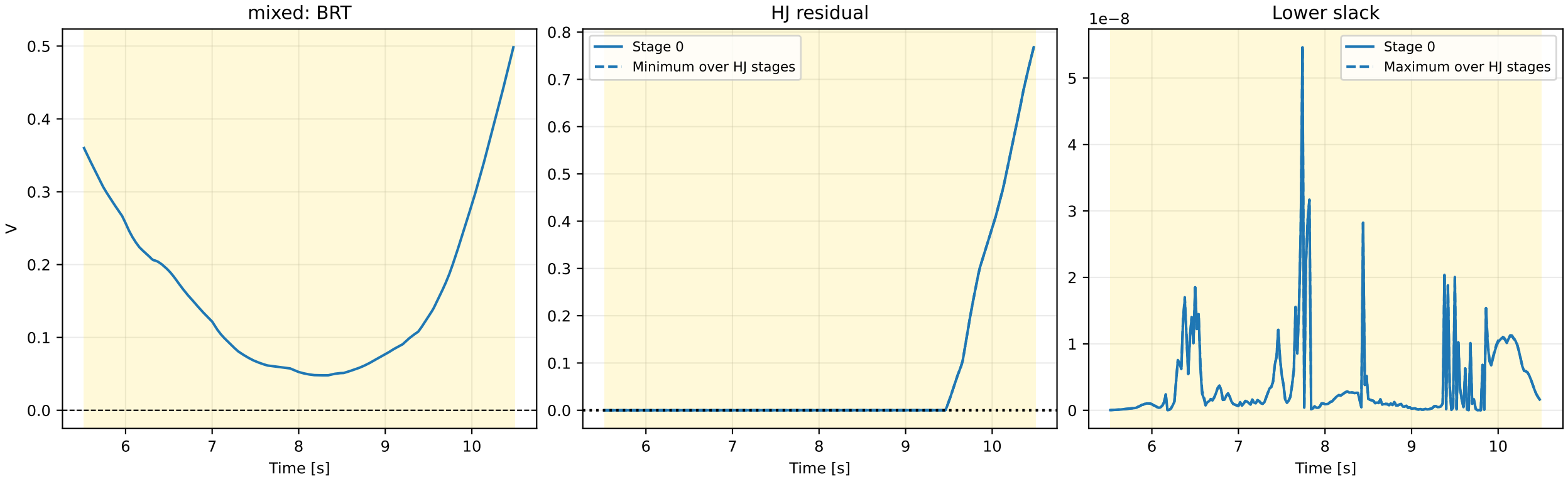
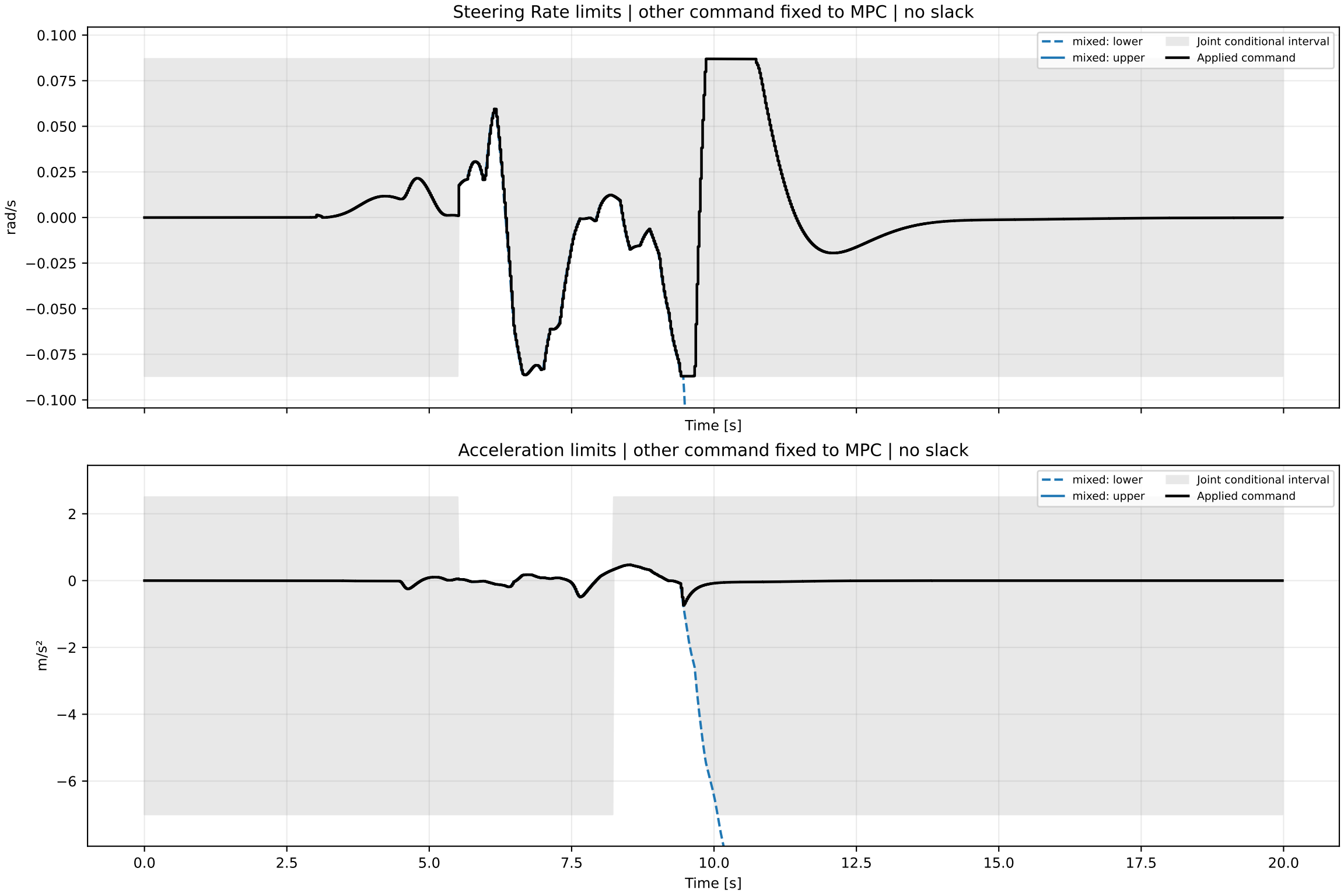
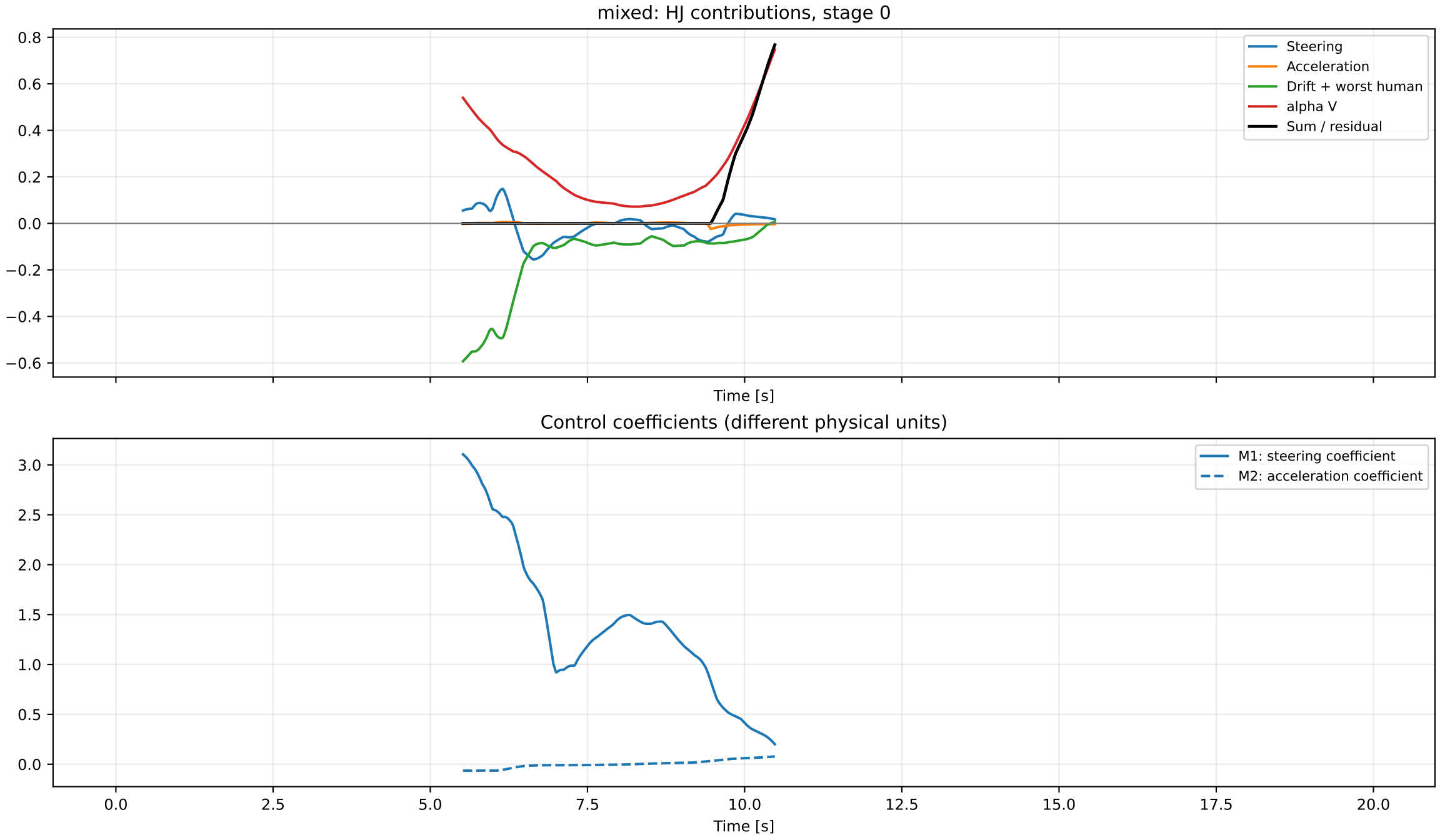
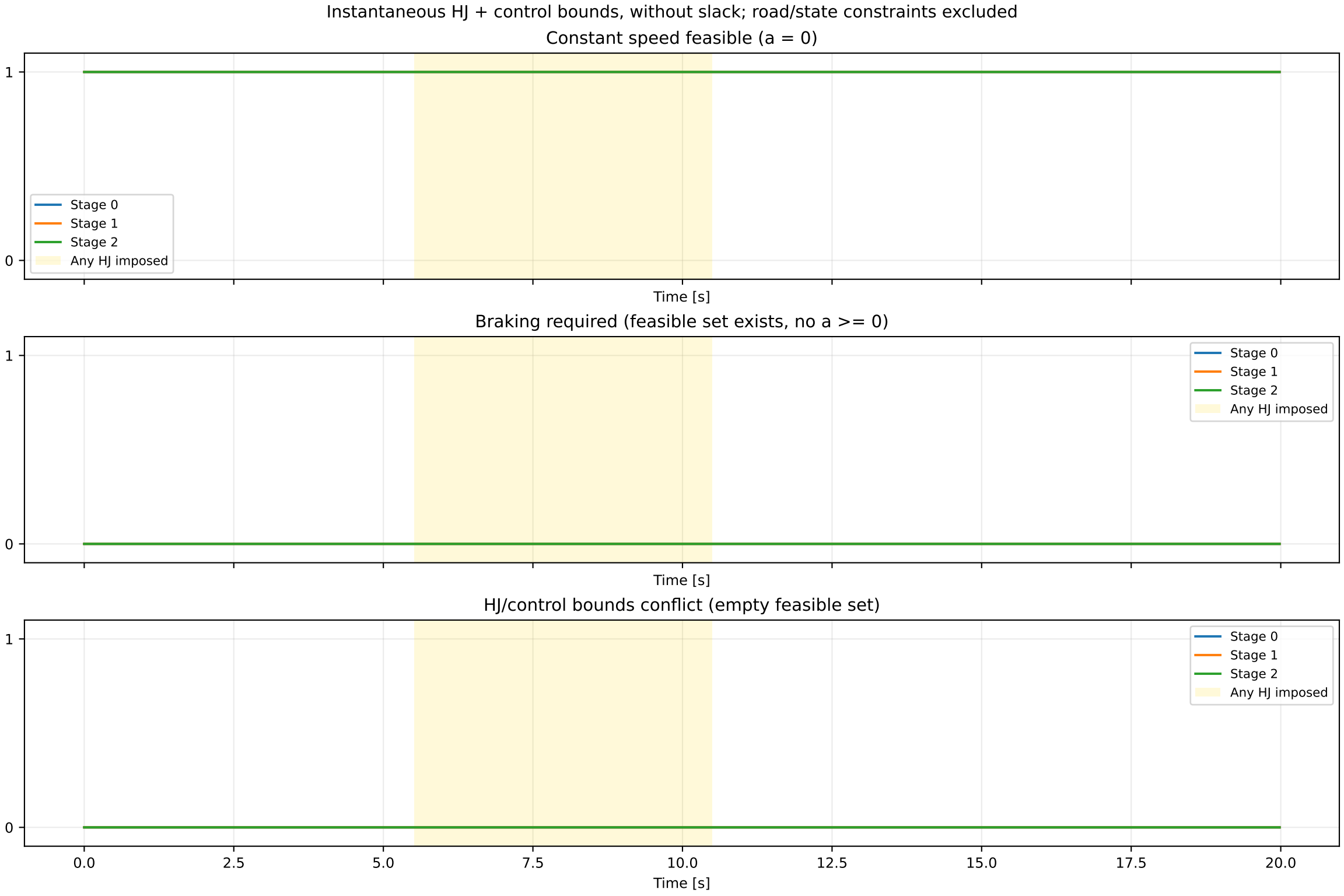
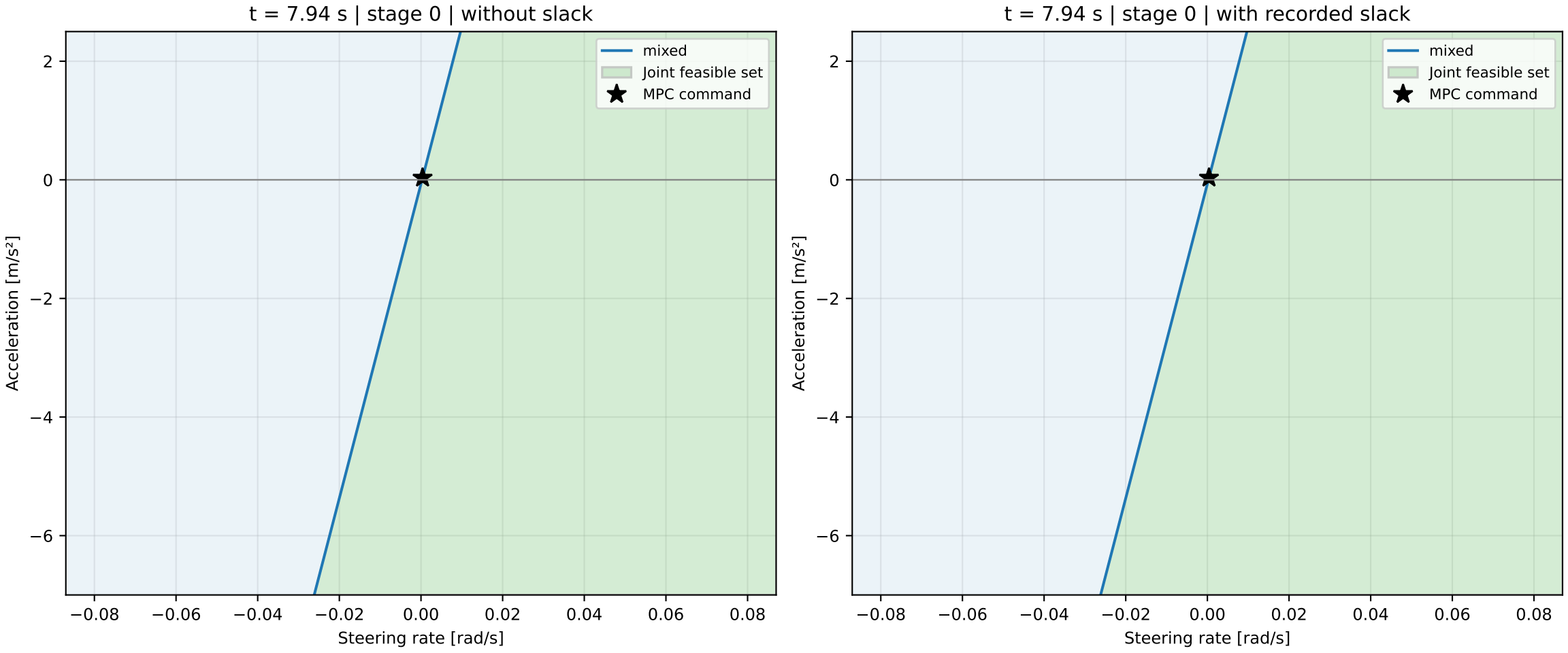

Global trajectory
Ego
Y [m]
X [m]
Velocity
Ego
Reference
m/s
Time [s]
Steering rate
rad/s
Time [s]
Acceleration
m/s²
Time [s]
Heading
deg
Time [s]
Steering angle
deg
Time [s]

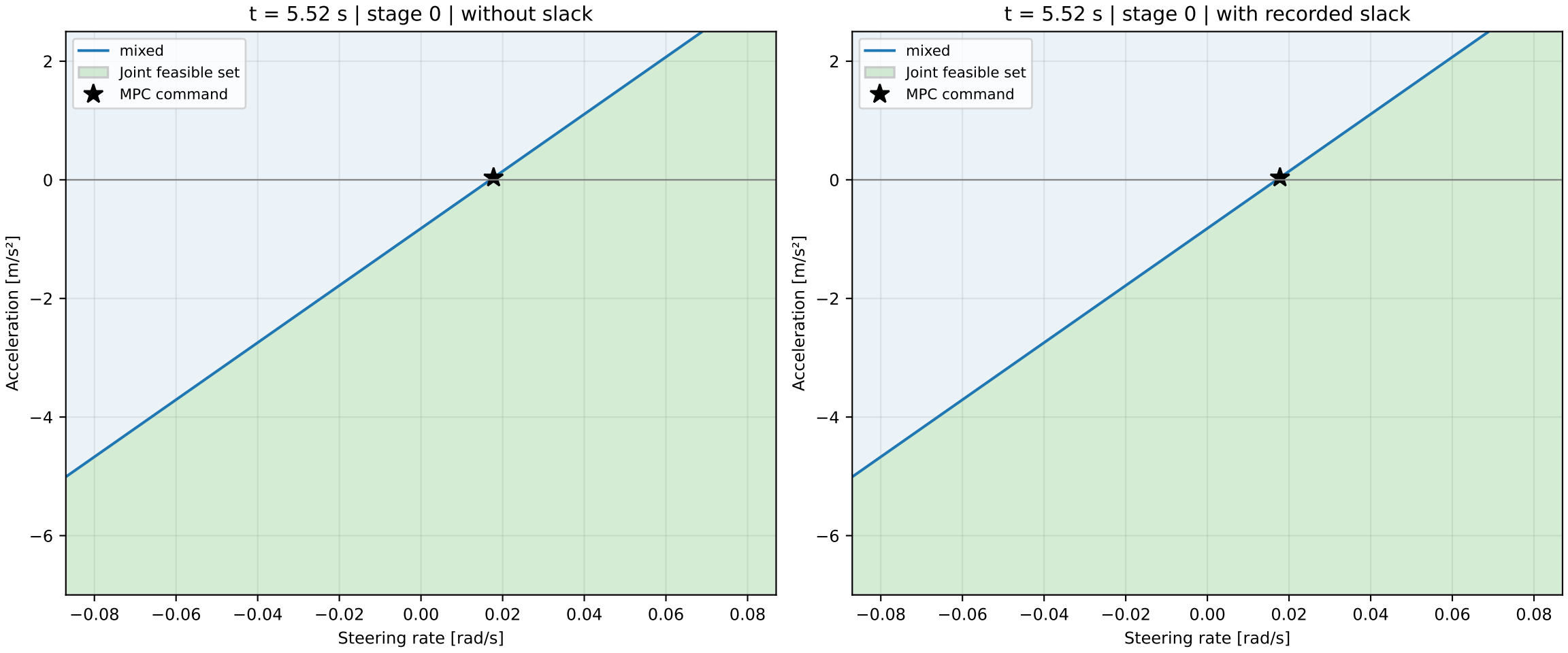

mixed: BRT
V
Time [s]
HJ residual
Stage 0
Minimum over HJ stages
Time [s]
Lower slack
1e−8
Stage 0
Maximum over HJ stages
Time [s]

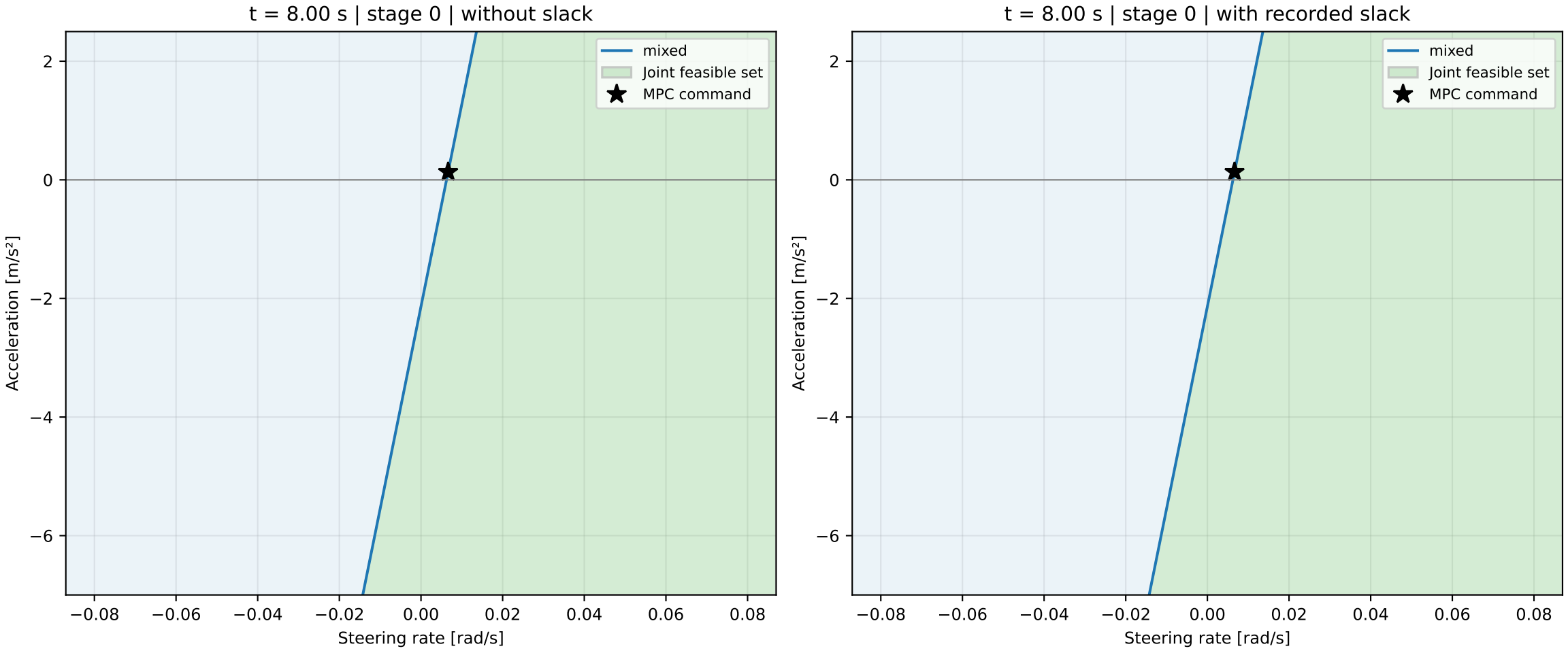

Steering Rate limits | other command fixed to MPC | no slack
mixed: lower
mixed: upper
Joint conditional interval
Applied command
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
rad/s
Time [s]
Acceleration limits | other command fixed to MPC | no slack
mixed: lower
mixed: upper
Joint conditional interval
Applied command
2
0
−2
−4
−6
m/s²
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

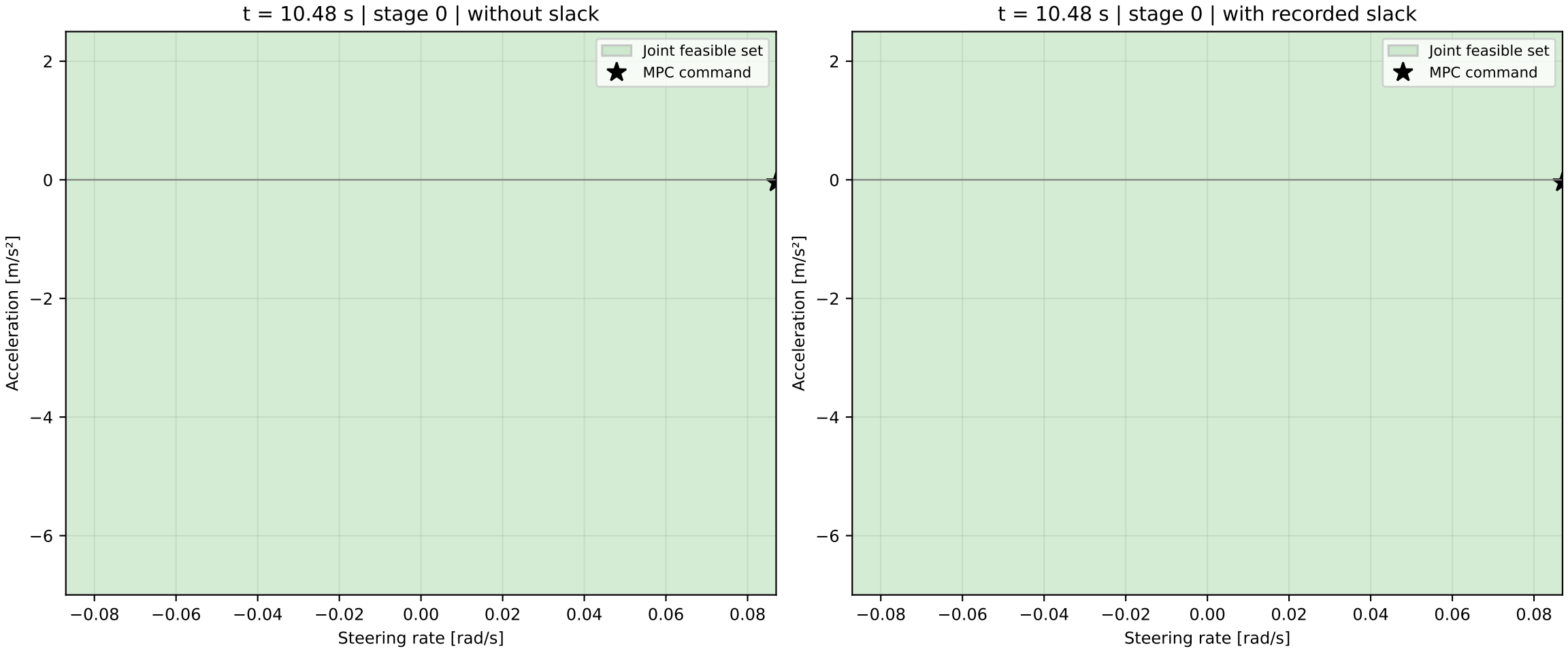

mixed: HJ contributions, stage 0
Steering
Acceleration
Drift + worst human
alpha V
Sum / residual
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Time [s]
Control coefficients (different physical units)
M1: steering coefficient
M2: acceleration coefficient
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

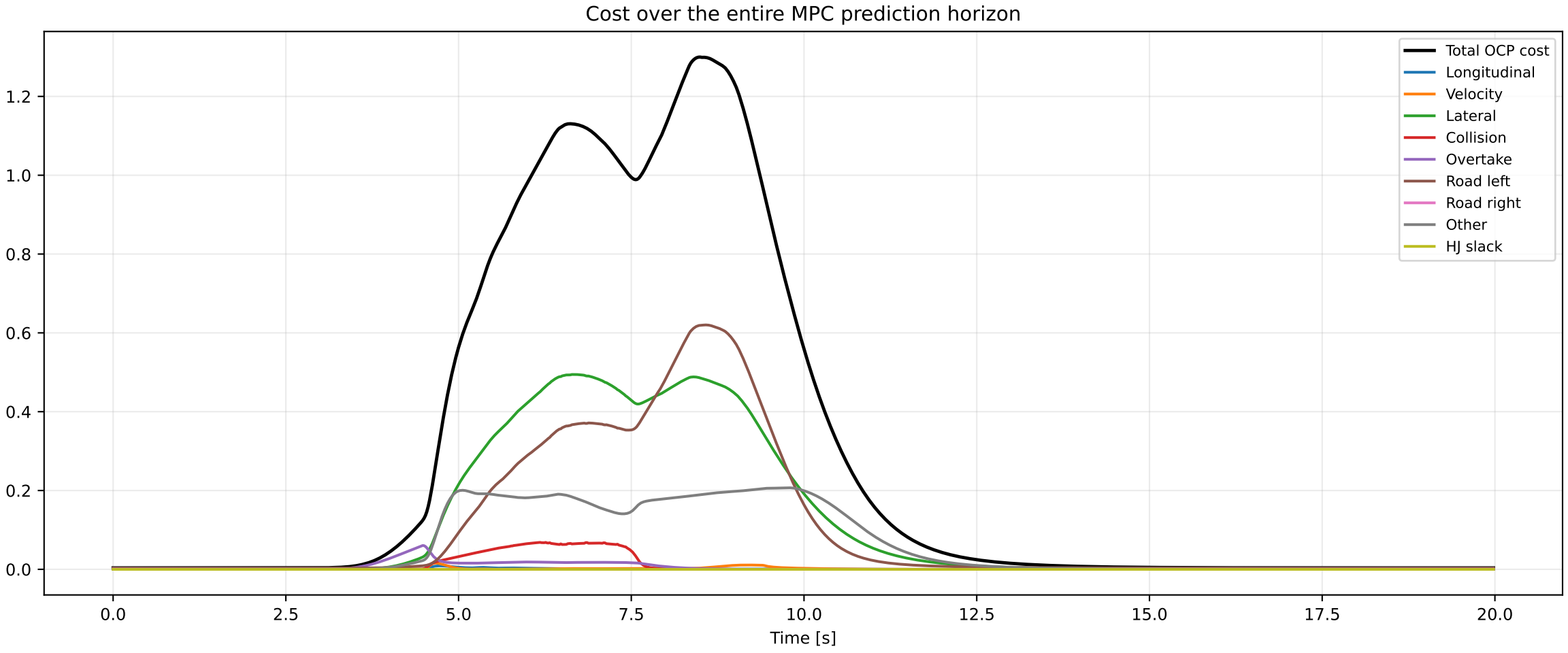

Instantaneous HJ + control bounds, without slack; road/state constraints excluded
Constant speed feasible (a = 0)
1
0
Stage 0
Stage 1
Stage 2
Any HJ imposed
Time [s]
Braking required (feasible set exists, no a >= 0)
1
0
Stage 0
Stage 1
Stage 2
Any HJ imposed
Time [s]
HJ/control bounds conflict (empty feasible set)
1
0
Stage 0
Stage 1
Stage 2
Any HJ imposed
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


t = 7.94 s | stage 0 | without slack
t = 7.94 s | stage 0 | with recorded slack
mixed
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


t = 5.52 s | stage 0 | without slack
t = 5.52 s | stage 0 | with recorded slack
mixed
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


t = 8.00 s | stage 0 | without slack
t = 8.00 s | stage 0 | with recorded slack
mixed
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


t = 10.48 s | stage 0 | without slack
t = 10.48 s | stage 0 | with recorded slack
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


Cost over the entire MPC prediction horizon
Total OCP cost
Longitudinal
Velocity
Lateral
Collision
Overtake
Road left
Road right
Other
HJ slack
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]